

The product-kernel block with unequal exponents

Claude, with Daniel Murfet

2026-09-07

Abstract

For $p_1 = (h_1 + 1)/k_1 < p_2 = (h_2 + 1)/k_2$ and a nonnegative kernel g , the product-kernel block $J(t) = \iint_{(0,b]^2} u^{h_1} v^{h_2} g(tu^{k_1} v^{k_2})$ is bounded by the smaller exponent alone (Astra #5(d)):

$$J(t) \leq \frac{b^{k_2(p_2-p_1)}}{k_1 k_2 (p_2 - p_1)} M_{p_1}(g) t^{-p_1}, \quad t > 0.$$

The density of unit 103 with constant amplitude is $k_2^{-1} r^{p_2-1} g(tr)(\varepsilon^{-\gamma} - b^{-\gamma})/\gamma$ with $\gamma = k_1(p_2 - p_1)$ and $\varepsilon = (r/b^{k_2})^{1/k_1}$, which is at most $b^{k_2(p_2-p_1)}/(k_1 k_2 (p_2 - p_1)) \cdot r^{p_1-1} g(tr)$. Together with the equal-exponent bound of unit 111 this makes the mixed remainder of the $d = 2$ cutoff theorem fully quantitative in both regimes. Zero `sorry/axiom`.

1 The things formalised

```
theorem genDensity_const_le (b p p : ℝ) (h k k : ℝ) (hb : 0 < b) (hk : 0 < k)
  (hk : 0 < k) (hp : ((h : ℝ) + 1) / k = p) (hlt : p < p) (g : ℝ → ℝ)
  (hg0 : s, 0 ≤ g s) (r s : ℝ) (hr : 0 < r) (hrR : r ≤ b ^ (k + k)) :
  genDensity p b h k k (fun _ s => g s) r s
  ≤ b ^ ((k : ℝ) * (p - p)) / ((k : ℝ) * k * (p - p)) * (r ^ (p - 1) * g s)
theorem prodKernel_le_unequal (b p p : ℝ) (h h k k : ℝ) (hb : 0 < b) (hk : 0 < k)
  (hk : 0 < k) (hp : ((h : ℝ) + 1) / k = p) (hp : ((h : ℝ) + 1) / k = p)
  (hlt : p < p) (g : ℝ → ℝ) (hg : Continuous g) (hg0 : s, 0 ≤ g s)
  (hM : IntegrableOn (fun s => s ^ (p - 1) * g s) (Ioi 0)) (t : ℝ) (ht : 0 < t) :
  prodKernel b h h k k g t
  ≤ b ^ ((k : ℝ) * (p - p)) / ((k : ℝ) * k * (p - p)) * kernelMoment p g
  * t ^ (-p)
```

2 Mathematics

With weight exponent $w = h_1 - k_1 p_2$, $w + 1 = -k_1(p_2 - p_1) = -\gamma < 0$, so $\int_\varepsilon^b u^w du = (\varepsilon^{-\gamma} - b^{-\gamma})/\gamma \leq \varepsilon^{-\gamma}/\gamma$ and $\varepsilon^{-\gamma} = (r/b^{k_2})^{-(p_2-p_1)} = r^{-(p_2-p_1)} b^{k_2(p_2-p_1)}$; the powers of r combine to $r^{p_2-1-(p_2-p_1)} = r^{p_1-1}$. The block integral is the r -integral of the density over $(0, R]$ (unit 103 at $\beta = 0$); `integral_mono_of_nonneg` compares it with the dominating integrand without needing integrability of the density itself (only nonnegativity and integrability of the bound, which follows from boundedness of g on $[0, tR]$), the substitution $x = tr$ produces t^{-p_1} , and the resulting moment over $(0, tR]$ is at most M_{p_1} .

3 Paper-facing meaning

The mixed remainder of the rectangular decomposition has shifted exponents $(h_1 + M_1 + 1)/k_1$ and $(h_2 + M_2 + 1)/k_2$, which coincide only when $M_1/k_1 = M_2/k_2$; the present bound covers the generic unequal case with an explicit constant, the unit-111 bound the equal case. The $1/(p_2 - p_1)$ factor is the artificial blow-up Astra #5(d) warned about near a collision; the gap-uniform alternative $(1 - x^\delta)/\delta \leq \log(1/x)$ remains a follow-up.

4 Lean notes

- `integral_mono_of_nonneg` needs integrability of the *dominating* function only; use it whenever the smaller integrand is awkward to show integrable.
- Power identities: after `Real.div_rpow`, `← Real.rpow_natCast`, `← Real.rpow_mul`, `div_eq_mul_inv` and `← Real.rpow_neg`, finish with `congr 2`; `ring` on the exponent; `rw` cannot see through $(b \wedge x)^{-1} = b \wedge (-x)$ in the other direction once other rewrites have fired.
- `div_neg`, `← neg_div`, `neg_sub` in that order rewrite $(a - c)/(-d)$ to $(c - a)/d$.
- `field_simp`; try `ring` accommodates both outcomes (closed or not).